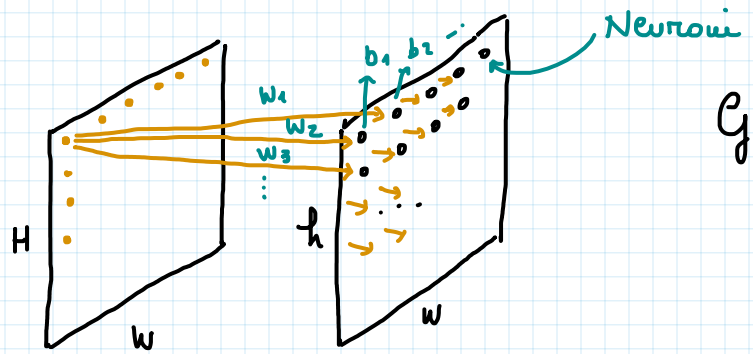
CNN

Def Si dice CORRELAZIONE INCROCIATA (CROSS-CORRELATION) di un segnale in INPUT $x\ \mathbb{R}\to\mathbb{R}$ contro $w:\mathbb{R}\to\mathbb{R}$ (KERNEL):

$$(x \star w)(t) = \int_{\mathbb{R}} x(a)\, w(t+a)\, da = (x * \underbrace{w^-}_{\text{KERNEL FLIPPATO}})(-t)$$

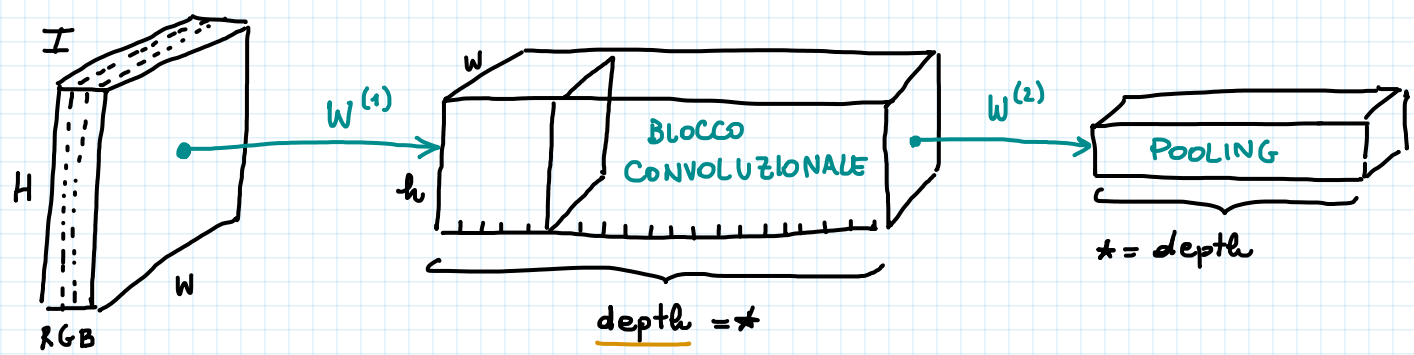
KERNEL FLIPPATO: $w^-(t) = w(-t)$

RETI NEURALI CONVOLUZIONALI

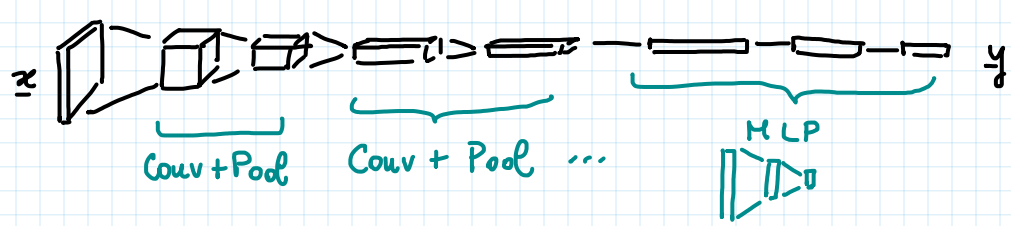


Gestire dati a "GRIGLIA"

Ragionamento a VOLUMI $\rightsquigarrow$ TENSORI



Struttura o ARCHITETTURA standard di una CNN:



INPUT

$$I \in \mathbb{R}^{H\times W\times \#channel} \xrightarrow{W^{(1)},\, b^{(1)}} C \in \mathbb{R}^{h_1\times w_1\times \#filtri} \xrightarrow{W^{(2)},\, b^{(2)}} P \in \mathbb{R}^{h_2\times w_2\times \#filtri}$$